Answer Key

Question 1
HPR = (Ending value − Beginning value + Dividends) / Beginning value
HPR = (52,000 − 50,000 + 2,000) / 50,000 = 4,000 / 50,000 = 8%

Question 2
HPR = (545,000 − 500,000 + 15,000) / 500,000 = 60,000 / 500,000 = 12%

Question 3
a) Arithmetic average = (20% + (−10%)) / 2 = 5%

b) Geometric average = $[(1.20)(0.90)]^{(1/2)} - 1 \approx 3.92\%$

The geometric average is lower because it accounts for compounding: the 10% loss in Year 2 applies to a base that already includes Year 1's gain, so it erases more dollars than the arithmetic average implies.

Question 4
A loss and the gain needed to recover it are measured against different starting values. A 10% loss on $100 leaves $90. To get back to $100 from $90 requires a gain of $10, but $10 is 11.1% of $90, not 10%. The recovery percentage is always calculated on a smaller base than the original loss, so it must be proportionally larger. The deeper the loss, the more pronounced this effect becomes (a 50% loss requires a 100% gain to recover).